COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions — every state change is recorded, so each order's history is auditable through ledger entries and notifications — and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers record pickup and delivery; administrators report. Deployment is local-only: the app runs on a single host for development and demo; customers, staff, drivers and administrators are roles in the same app (§8). Stock checks, one confirmation transaction (§5) and role gates (FR-07) guard integrity.

2. Development process

An increment is a unit of work closed by a review or a recorded test run; a date range spans the work.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Coursework kickoff	Scope, responsibilities and contracts — agreed endpoint and data shapes — recorded (increment register)
17 Oct 2026 review	Order and stock functions ready for review	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a confirmation endpoint that accepted repeat submits and a missing ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Order, stock, driver and admin functions complete	Application integrated; all four role workflows runnable end to end
Driver and privacy, 2 Nov 2026	Driver assignment tests and data-minimisation checks outstanding	Driver assignment and data-minimisation checks pass (tests)
Diagram and traceability review, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (diagram review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three in-app interface interactions; the 11 Nov regression exercised all four role workflows (browser checks)
CI health tests, 11–12 Nov 2026	The record of CI-run outcomes (observations) had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements

FR	The system shall	Verification
FR-01	create orders with one or more items and totals	Test
FR-02	enforce lifecycle and stock reservation/release	Test (cancel tests)
FR-03	number orders and shipments and show owner tracking (the staff member or driver currently holding each order)	Test
FR-04	show inventory, replenishment, stock-movement ledger and low-stock alerts	Test
FR-05	record notifications for confirm, ship, pickup, delivery and cancel, and allow reading	Test
FR-06	provide admin aggregates and CSV export	Test
FR-07	authenticate and enforce role/object access	Test
FR-08	assign shipped orders and record delivery	Test

- NFR-01 performance — API p95 ≤2 s, first contentful paint p95 ≤3 s (measurement, §6). - NFR-02 CI health — green build, test and browser runs, recorded per CI run (observations). - NFR-03 least privilege — role gates on mutating endpoints; staff ship through order-state changes and the delivery address is released only to the assigned driver (privacy tests). - NFR-04 interaction budget — at most three in-app interface interactions per core operation (the eleven listed in §8); one interaction is a fill, selection or click on an in-app control, and login/navigation are setup (walkthrough). - NFR-05 process quality — green CI runs per increment (observations).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	scheduling plan	increment register
RK-02 contract rework	Transition table locked by constant tests	transition tests
RK-03 test-week capacity	Traceability	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed data regression-tested	seed regression test
RK-06 coursework assignments and test-week squeeze	Weekly re-planning	risk plan
RK-07 stack unfamiliarity	Setup guide and CI	setup guide
RK-08 report and presentation-video deadlines	Weekly re-planning	risk plan

4. System models

The four layers — role pages, HTTP routes, business handlers, Prisma/SQLite persistence (ORM over the SQLite database) — separate interface, application and data concerns. Figures 1–5 answer, in order.

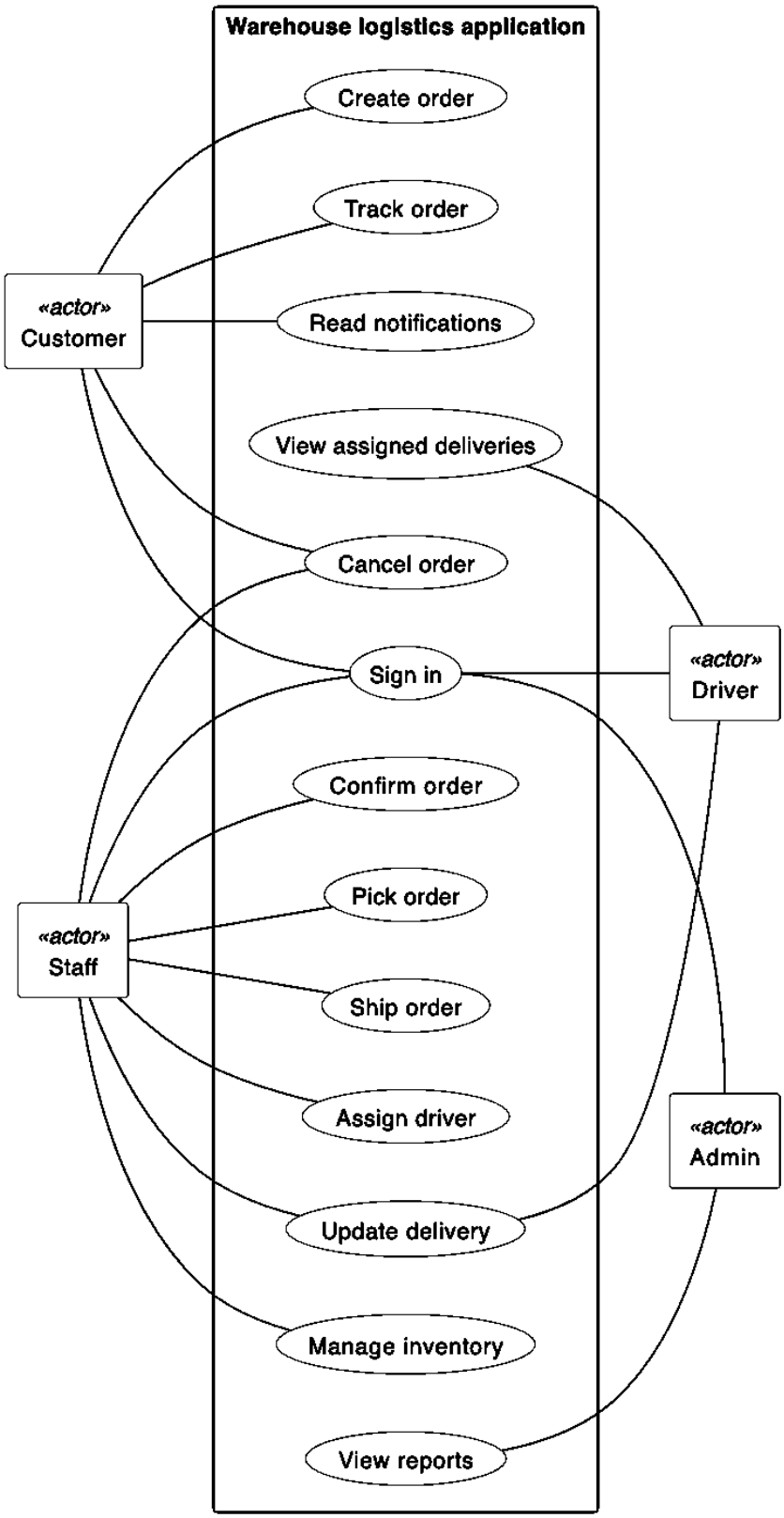


Figure 1. Role functions: thirteen use cases grouped under four role actors. Source.

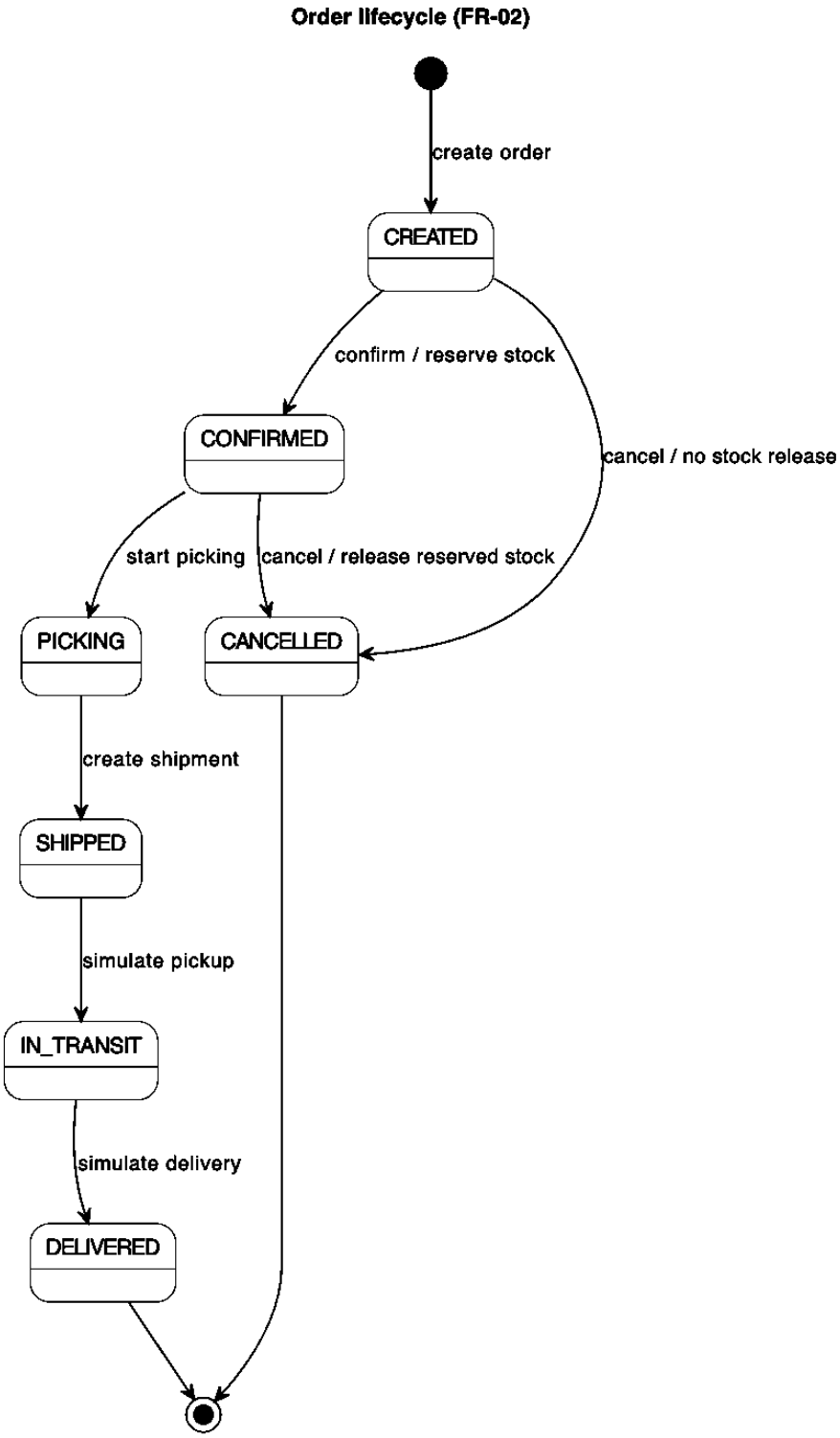


Figure 2. Order transitions and stock effects: each transition states its stock effect — reserve, release or none. Source.

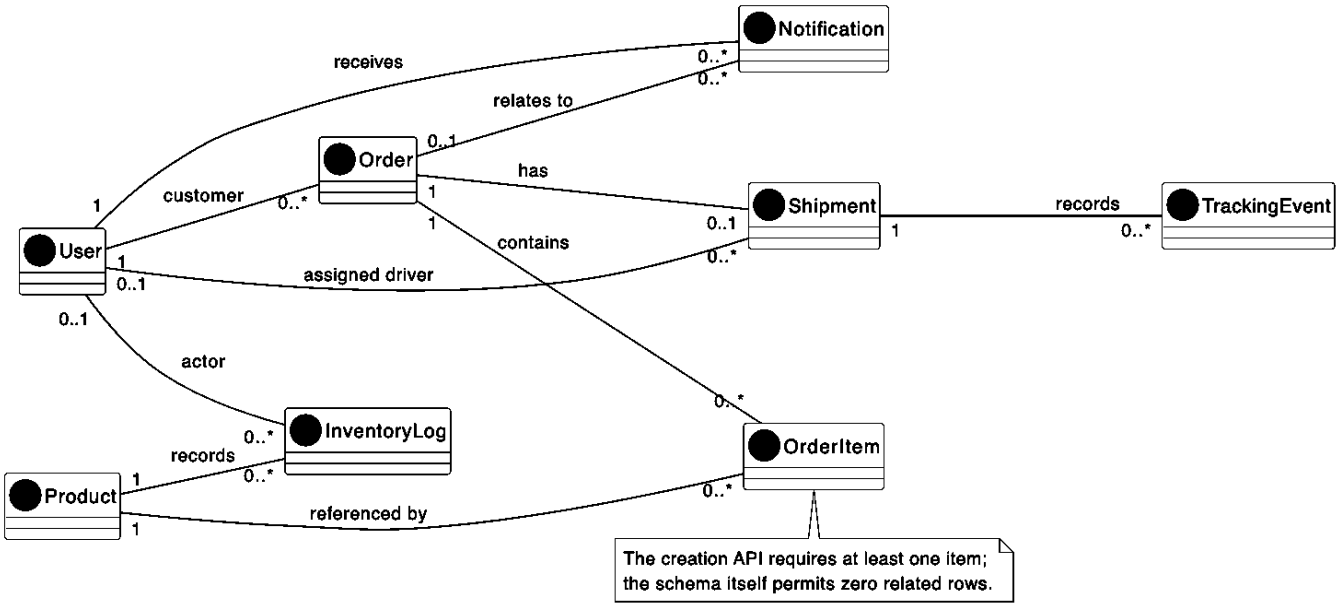


Figure 3. Data relationships. Source.

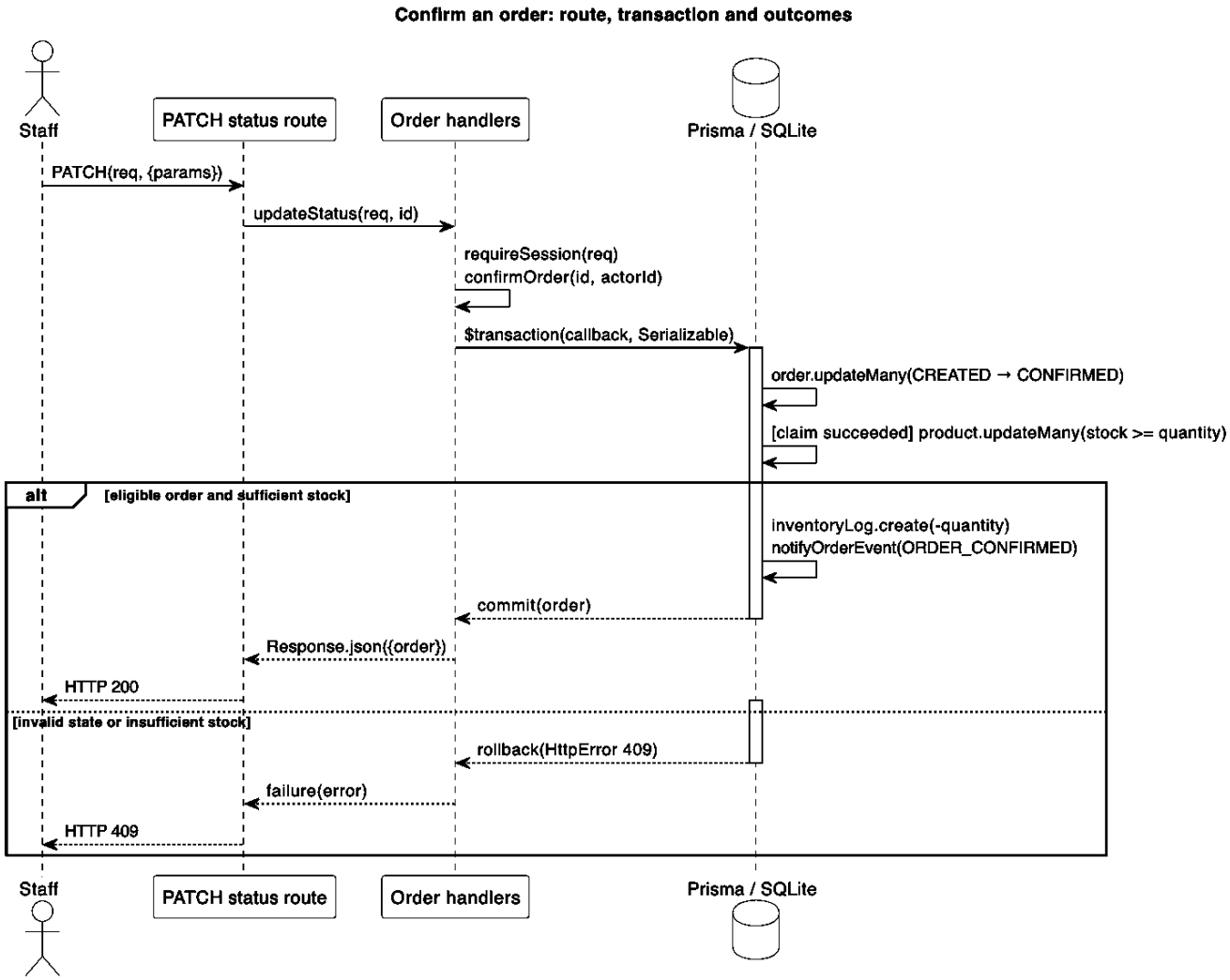


Figure 4. Confirmation transaction: session guard, conditional writes, rollback on failure or short stock. Source.

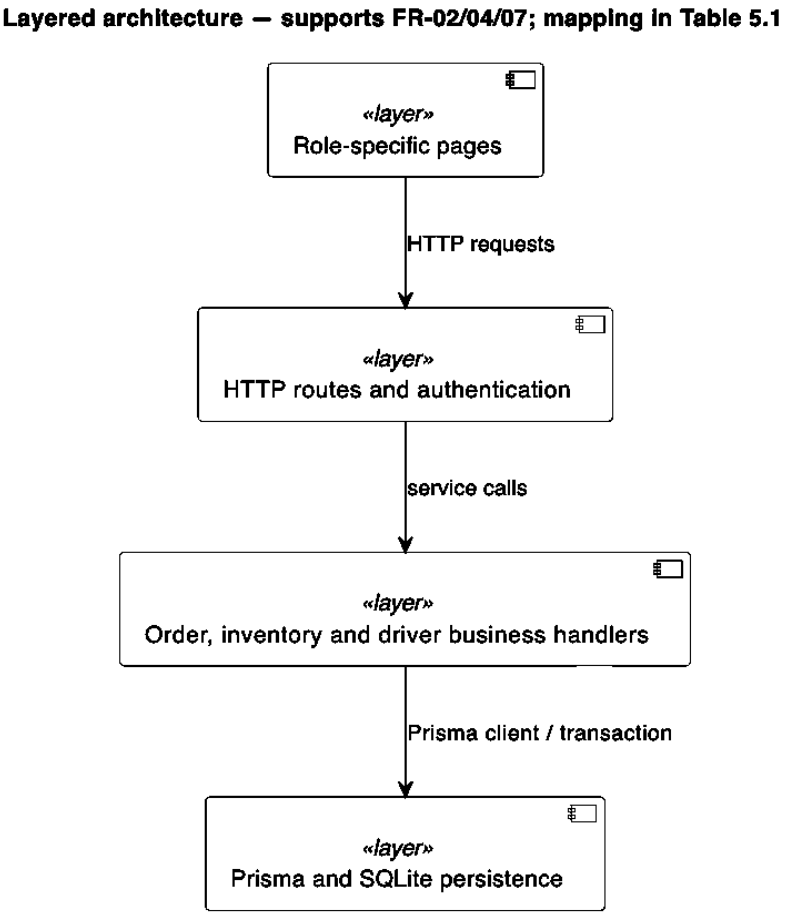


Figure 5. Four layers. Source.

5. Implementation

The confirmation transaction (order-api.ts:245–293) reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED from CREATED or CONFIRMED. Cancellations restock unless the order was CREATED. Staff cancellation is restricted to CONFIRMED; a CREATED order is still the customer's to cancel.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED, sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: writes state, stock, ledger and notifications; rolls back on insufficient stock; repeats rejected (409)	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 ledger	order-api.ts:281, 398	Inventory
FR-05/06/07 handlers	notify.ts:19, reports-api.ts:105, http.ts:58 — follow the same handler pattern (separate modules)	Notifications, Reports, Auth

6. Verification

CI records 38 unit and 140 integration checks; 2 integrity checks — the pipeline refuses to run without an explicit confirmation flag (`RESET_DEMO_DATA=YES`), and the packaged bundle digest must match the verified source (`SHA256SUMS.txt`) (integrity checks); the integration log "tests 140 / pass 140", the regression log "Tests 38 passed (38)".

Loopback measurement (11 Nov; 30 requests per endpoint across 6 endpoints) gave API p95 3.2–13.6 ms and first-contentful-paint p95 64–96 ms against 2 s and 3 s targets — loopback only, not representative of public networks or high concurrency.

The three recorded reviews (17 Oct, 10–11 Nov, 11 Nov) additionally served as static verification of requirements, models and code ([Table A](§2)).

7. Evolution

Three reviews drove the recorded changes: the 17 Oct review found the confirmation defect, fixed by the §5 transaction; the 10–11 Nov review aligned diagrams and traceability with the implementation and restated NFR-01/02 for local deployment; the 11 Nov review added the direct export path (issue #15) and two CI-health regression tests (suite 36 → 38).

8. Validation

The automated walkthrough (26 browser checks, recorded 11 Nov) exercises eleven core operations — Create an order; Open order details; Confirm order; Start picking; Ship order; Assign driver; Simulate pickup; Simulate delivery; Apply report date filter; Export current report; Export selected dates without applying filter — at most three in-app interface interactions each. "Browser acceptance passed: 26 checks" — 20 operation checks (one to 3 each), plus 6 cross-cutting checks (mobile viewport, dark scheme, sign-out).

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagrams diverged from the schema	Models corrected	10–11 Nov	Diagra
Stock effects were recorded under FR-04	Moved stock effects to FR-02	17 Oct	Traceal
Early register rows had thin triggers and outcomes	Register rewritten 10–11 Nov (early trigger cells held statuses, not events); earlier rows in git history	10–11 Nov	Registe
The filter-then-export path took four interactions	Selected-date export path added; issue closed	11 Nov walkthrough	Issue #
No regression test for CI health	Reached 38 tests	11 Nov regression work	Test his
NFR-02 originally required daily availability 08:00–24:00 local time	Replaced by CI-run checks (§3) — with local-only deployment (§1) there is no hosted environment whose daily window could be measured; each green run verifies the integration points per commit	10–11 Nov	Require

Planned: validation sessions with external participants are scheduled 17–24 Nov (§8); a performance re-probe follows on 25 Nov (§6).

10. Evidence

Baseline record — the suite counts at final packaging: 38 unit, 140 integration, 2 integrity; 26 browser checks recorded 11 Nov (baseline).

Traceability.